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Telecommunications Project Report

Simulation of a GNSS Transmission Chain

BPSK/QPSK Modulation, AWGN Channel and GNSS Effects

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1 Introduction

In this project, a digital communication chain inspired by a satellite navigation system (GNSS, *Global Navigation Satellite System*) is simulated. The objective is to model the end-to-end transmission of GNSS information (including latitude, longitude, altitude, and timestamp) from data encoding to reception and demodulation.

The transmission is performed using BPSK and QPSK digital modulation schemes over a channel impaired by additive white Gaussian noise (AWGN). In addition, GNSS-specific effects such as the Doppler frequency shift are introduced in order to analyze their impact on the received signal, constellation diagrams, and symbol error rate (SER).

1.1 General System Architecture

We consider the classical transmission chain represented by the following diagram:

$$\boxed{\text{Transmitter}} \rightarrow \boxed{\text{GNSS Channel}} \rightarrow \boxed{\text{Receiver}}$$

1.2 Project Objectives

The main objectives of this work are as follows:

1. Encode GNSS data in bit form, then as BPSK/QPSK symbols.
2. Model a noisy AWGN channel with different signal-to-noise ratios (SNR).
3. Implement a receiver based on matched filtering and coherent demodulation.
4. Analyze performance in terms of received constellation and symbol error rate (SER).
5. Study the impact of a GNSS-specific effect, such as Doppler effect.

2 Description of the Transmitted Signal

2.1 GNSS Data and Binary Encoding

The information transmitted in our system consists of: latitude lat (in degrees), longitude lon (in degrees), altitude alt (in meters), and date, represented as the number of seconds elapsed since January 1st, 1970 (*Unix timestamp*).

These quantities are first scaled and then converted to integers according to the following relations:

$$\text{lat_int} = \lfloor \text{lat} \times 10^7 \rfloor, \tag{1}$$

$$\text{lon_int} = \lfloor \text{lon} \times 10^7 \rfloor, \tag{2}$$

$$\text{alt_int} = \lfloor \text{alt} \times 100 \rfloor. \tag{3}$$

Each integer is then converted into a fixed-size bit vector (for example 32 bits for latitude, 32 bits for longitude, 16 bits for altitude and 32 bits for date), via a function of the type:

$$\text{int_to_bits}(n, L) \in \{0, 1\}^L. \tag{4}$$

The total transmitted binary vector is therefore the concatenation:

$$\mathbf{b} = \left[\mathbf{b}_{\text{lat}}, \mathbf{b}_{\text{lon}}, \mathbf{b}_{\text{alt}}, \mathbf{b}_{\text{time}}\right], \tag{5}$$

where each sub-vector corresponds to a physical quantity.

2.2 BPSK Modulation

In BPSK (*Binary Phase Shift Keying*), each bit is associated with a real symbol according to the rule:

$$b_k \in \{0, 1\} \quad \Rightarrow \quad s_k = 2b_k - 1 \in \{-1, +1\}. \tag{6}$$

The baseband signal can then be written as:

$$x_{\text{BPSK}}(t) = \sum_k s_k \, p(t - kT_s), \tag{7}$$

where $p(t)$ is the shaping pulse (for example rectangular) and T_s is the symbol duration.

2.3 QPSK Modulation

In QPSK (*Quadrature Phase Shift Keying*), bits are grouped in pairs $(b_{2k}, b_{2k+1}) \in \{0,1\}^2$. These pairs are then mapped onto normalized complex symbols according to the Gray constellation:

$$\begin{cases} (0,0) \mapsto \dfrac{1}{\sqrt{2}}(1+j), \\ (0,1) \mapsto \dfrac{1}{\sqrt{2}}(-1+j), \\ (1,1) \mapsto \dfrac{1}{\sqrt{2}}(-1-j), \\ (1,0) \mapsto \dfrac{1}{\sqrt{2}}(1-j). \end{cases} \tag{8}$$

We then obtain a complex baseband signal:

$$\tilde{x}(t) = x_I(t) + jx_Q(t) = \sum_k a_k\, p(t - kT_s), \tag{9}$$

where a_k are the QPSK symbols in the IQ plane.

At carrier frequency f_c, the transmitted signal is written as:

$$x_T(t) = x_I(t)\cos(2\pi f_c t) - x_Q(t)\sin(2\pi f_c t). \tag{10}$$

3 GNSS Channel Modeling

3.1 Additive White Gaussian Noise (AWGN)

The channel is modeled by additive white Gaussian noise with zero mean:

$$y(t) = x_T(t) + n(t), \tag{11}$$

where $n(t)$ is a wide-sense stationary Gaussian process, with power spectral density $S_n(f) = N_0/2$.

In discrete form, this corresponds to:

$$y_k = s_k + n_k, \tag{12}$$

where $n_k \sim \mathcal{N}(0, \sigma^2)$, with σ^2 depending on the chosen SNR.

The signal-to-noise ratio (SNR) is defined in decibels as:

$$\mathrm{SNR}_{\mathrm{dB}} = 10 \log_{10} \left(\frac{E_s}{N_0} \right), \tag{13}$$

where E_s is the symbol energy. In the simulation, different SNR values will be tested (for example 0 dB and 10 dB) to illustrate the impact of noise.

3.2 Simplified Power Link Budget

In order to provide a physical interpretation of the simulated (E_b/N_0) values, a simplified power link budget is considered for a GNSS-like scenario.

We assume a LEO satellite at an altitude of 350 km transmitting at a carrier frequency $f_c = 30\,\mathrm{MHz}$. The transmitted power is $P_t = 10\,\mathrm{W}$ with a satellite antenna gain $G_t = 10\,\mathrm{dBi}$ and a receiver antenna gain $G_r = 0\,\mathrm{dBi}$. Additional system losses are estimated to be $L_{\mathrm{sys}} = 3\,\mathrm{dB}$.

The free-space path loss is given by:

$$L_{\mathrm{fs}} = 32.44 + 20 \log_{10}(f_{\mathrm{MHz}}) + 20 \log_{10}(d_{\mathrm{km}}), \tag{14}$$

which yields $L_{\mathrm{fs}} \approx 113\,\mathrm{dB}$.

The received power is therefore:

$$P_r = P_t + G_t + G_r - L_{\mathrm{fs}} - L_{\mathrm{sys}} \approx -96\,\mathrm{dBW}. \tag{15}$$

Assuming a system noise temperature of $T = 290\,\mathrm{K}$ and a bit rate $R_b = 1\,\mathrm{Mbps}$, the available energy-per-bit-to-noise ratio is:

$$\frac{E_b}{N_0} \approx 48\,\mathrm{dB}. \tag{16}$$

This value is significantly higher than the (E_b/N_0) used in the AWGN simulations (e.g., 10 dB), which confirms that the simulation operates in a normalized baseband framework and does not explicitly include propagation losses. The link budget nevertheless provides an order of magnitude consistent with GNSS systems and validates the choice of simulation parameters.

3.3 Additional GNSS Effect: Doppler Effect

To customize the simulation to the GNSS context, we can take into account the Doppler effect due to the relative satellite/receiver motion. This results in a frequency shift:

$$f_r = f_c + f_D, \tag{17}$$

where f_D is the Doppler frequency, a function of the relative velocity and carrier frequency.

In the complex baseband model, this effect can be modeled by a time-dependent phase rotation term:

$$\tilde{y}(t) = \tilde{x}(t)\, e^{j2\pi f_D t} + \tilde{n}(t). \tag{18}$$

4 Receiver and Signal Processing

4.1 Demodulation and Low-Pass Filtering

The receiver performs coherent demodulation by multiplying the received signal by local oscillators in phase and quadrature, then applies a low-pass filter to extract the baseband components $y_I(t)$ and $y_Q(t)$.

4.2 Matched Filter

To maximize the SNR at the sampling instant, a matched filter is used whose impulse response is:

$$h(t) = p^*(T_s - t), \tag{19}$$

where $p(t)$ is the shaping pulse.

In discrete form, the matched filter can be implemented by direct convolution or by Fast Fourier Transform (FFT). After filtering and sampling, decision samples $\hat{a}_k$ are obtained.

4.3 Decision and Bit Demodulation

In BPSK, the decision is simply based on the sign:

$$\hat{b}_k = \begin{cases} 1 & \text{if } \hat{s}_k \geq 0, \\ 0 & \text{otherwise.} \end{cases} \tag{20}$$

In QPSK, the decision is made separately on the real and imaginary components, thus determining the quadrant of the complex symbol $\hat{a}_k$ to recover bits $\hat{b}_{2k}$ and $\hat{b}_{2k+1}$.

4.4 Performance Evaluation

The symbol error probability (SER) estimated by simulation is defined as:

$$\widehat{P}_{\text{SER}} = \frac{\text{number of erroneous symbols}}{\text{total number of symbols}}. \tag{21}$$

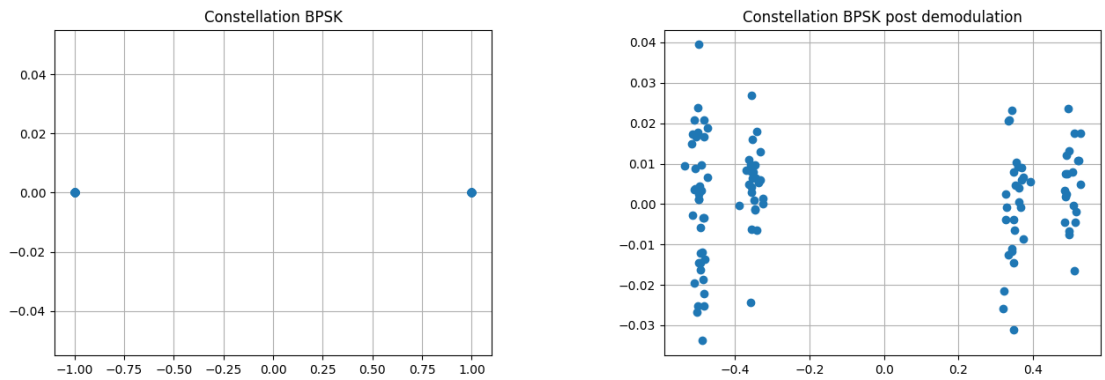
This estimate will be compared with theoretical expressions for BPSK and QPSK over AWGN channel.

5 Simulation Results

In this section, we present the results obtained with the Python implementation of the transmission chain.

5.1 BPSK Constellation

Figure 1 presents the BPSK constellation diagram. The ideal constellation consists of two points on the in-phase axis. After transmission over an AWGN channel and demodulation, the received symbols spread around these ideal positions. As expected, the dispersion decreases as the signal-to-noise ratio (SNR) increases.



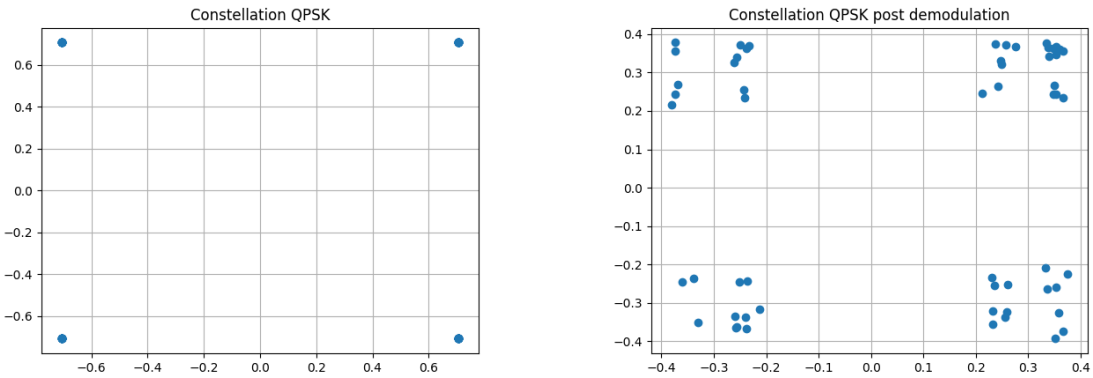
(a) Ideal BPSK constellation.

(b) BPSK constellation after demodulation.

Figure 1: BPSK constellation diagrams (ideal vs. noisy case).

5.2 QPSK Constellation

Figure 2 shows the QPSK constellation. In the ideal case, the symbols are located at four points in the complex plane. With AWGN and after demodulation, the received symbols form clusters around these positions. Similarly to BPSK, the clusters become tighter as the SNR increases.



(a) Ideal QPSK constellation. (b) QPSK constellation after demodulation.

Figure 2: QPSK constellation diagrams (ideal vs. noisy case).

5.3 Time Domain Signal Representation

The time-domain representation of the transmitted baseband signal highlights the structure of the modulation process. The in-phase (I) and quadrature (Q) components are shown separately, allowing a clear visualization of the symbol sequence and pulse shaping.

In this case, a rectangular pulse shaping is used, resulting in piecewise constant waveforms with abrupt transitions between symbols. This representation provides insight into the temporal behavior of the modulated signal before transmission over the channel.

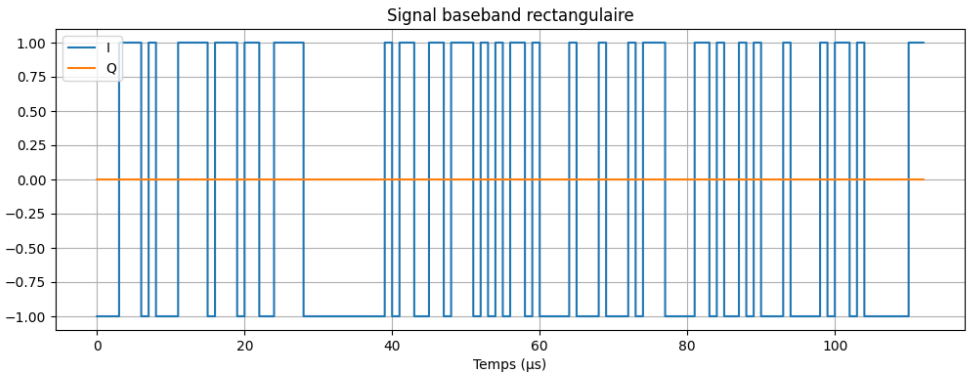


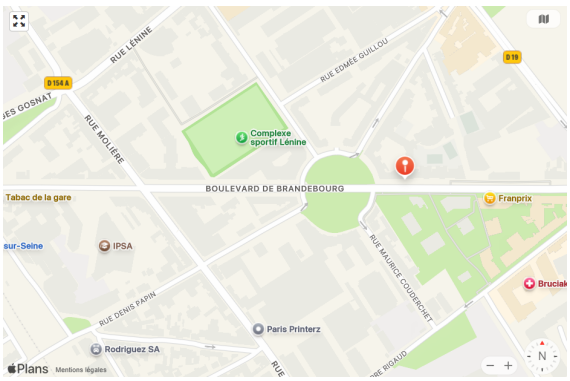
Figure 3: Time-domain representation of the baseband signal showing the in-phase (I) and quadrature (Q) components with rectangular pulse shaping.

5.4 Numerical Results at Fixed SNR

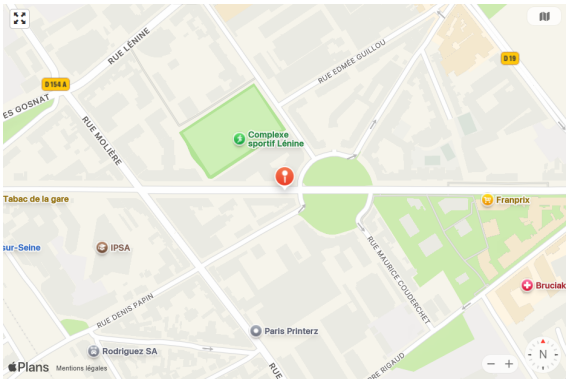
In order to illustrate the impact of the signal-to-noise ratio on system performance, simulations were performed for two representative SNR values: 5 dB and 10 dB. For each case, the Symbol Error Rate (SER) was evaluated and the reconstructed GNSS parameters were compared with the transmitted values. Additional time-domain and frequency-domain plots were generated during the simulation to validate the implementation but are not shown here for clarity.

Table 1: Simulation results for different SNR values (BPSK modulation).

SNR (dB)	SER (simulation)	SER (theoretical)	GNSS data recovery
5	3.57×10^{-2}	5.95×10^{-3}	Partially incorrect
10	0	3.87×10^{-6}	Perfect recovery



(a) GNSS position reconstruction at 5 dB SNR.



(b) GNSS position reconstruction at 10 dB SNR.

Figure 4: Comparison of GNSS position reconstruction for two SNR values. A higher SNR significantly improves symbol detection and navigation accuracy.

At 5 dB, the simulated SER remains relatively high, leading to errors in the decoded GNSS parameters. While the latitude remains close to the transmitted value, noticeable errors appear in the longitude, altitude, and timestamp fields. This behavior is consistent with the theoretical expectations at low SNR, where noise significantly affects symbol decisions.

At 10 dB, no symbol errors were observed during the simulation, resulting in a perfect recovery of all transmitted GNSS data. The simulated SER is therefore equal to zero, while the theoretical SER remains very small but non-zero. This discrepancy is explained by the finite number of transmitted symbols in the simulation, which limits the observation of rare error events.

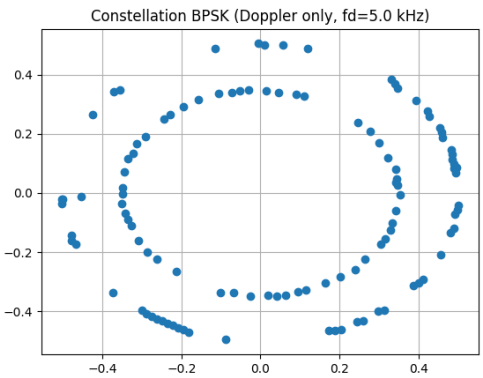
5.5 Doppler Effect Impact

The Doppler effect corresponds to a relative frequency shift between the transmitter and the receiver caused by their relative motion. In GNSS systems, this phenomenon introduces a residual frequency offset after down-conversion, which directly impacts the received baseband signal.

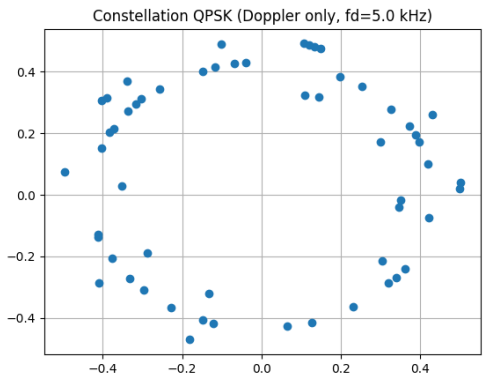
Doppler effect only (no noise)

When only the Doppler effect is considered, the residual frequency offset at baseband results in a continuous rotation of the received constellation. This behavior is observed for both BPSK and QPSK modulations, where the constellation points describe circular trajectories in the complex plane.

In this noise-free case, the constellations exhibit multiple circular bands instead of a single ideal circle. This effect is mainly due to the use of a rectangular pulse shaping, which introduces inter-symbol interference (ISI). Samples taken at different instants within the symbol period therefore experience slightly different amplitudes, leading to distinct circular trajectories under Doppler rotation.



(a) BPSK constellation with Doppler effect only.

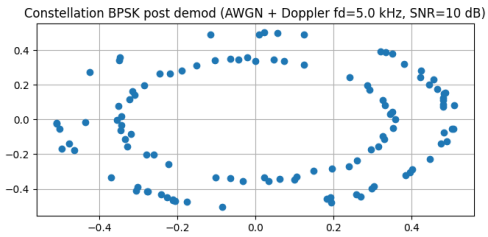


(b) QPSK constellation with Doppler effect only.

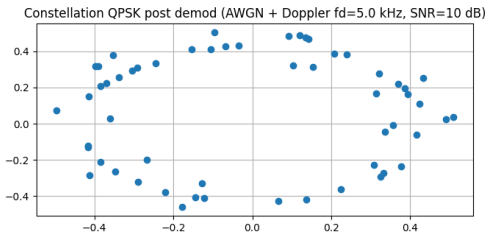
Figure 5: Constellation diagrams with Doppler effect only (no additive noise).

Combined Doppler effect and AWGN

When additive white Gaussian noise is introduced in addition to the Doppler effect, the constellation points become further dispersed around the rotating trajectories. The combined impact of noise and Doppler significantly increases the symbol error rate (SER), emphasizing the importance of Doppler compensation and synchronization techniques in practical GNSS receivers.



(a) BPSK constellation with Doppler effect and AWGN.



(b) QPSK constellation with Doppler effect and AWGN.

Figure 6: Constellation diagrams with combined Doppler effect and additive noise.

6 Conclusion

This project focused on the simulation of a digital communication chain inspired by a Global Navigation Satellite System (GNSS). Starting from realistic navigation data, an end-to-end transmission was implemented, covering data encoding, BPSK and QPSK modulation, channel impairments, and receiver processing. The study was conducted within a normalized baseband framework, allowing a clear analysis of the fundamental mechanisms governing digital satellite communications.

Through the simulation of an additive white Gaussian noise (AWGN) channel, the impact of noise on constellation diagrams and symbol error rate (SER) performance was investigated. The results obtained show a good agreement between theoretical and simulated SER values, validating the correctness of the implemented transmission chain. Furthermore, the reconstruction of GNSS parameters highlights the strong dependence of navigation accuracy on the signal-to-noise ratio, with reliable data recovery achieved only at sufficiently high SNR levels.

In addition to noise, the Doppler effect was introduced to reflect a key impairment specific to GNSS systems. The resulting constellation rotations clearly illustrate the impact of residual frequency offsets on coherent demodulation. The combined presence of Doppler shift and noise significantly degrades system performance, emphasizing the need for accurate synchronization and Doppler compensation in practical receivers.

Overall, this work provides a coherent and practical insight into the signal processing challenges involved in GNSS communication systems. The developed simulation framework constitutes a solid basis for further investigations, such as the use of more advanced pulse shaping techniques, error-correcting codes, or robust synchronization algorithms, which would allow a closer approximation to real-world GNSS receiver performance.